



AMITY UNIVERSITY

AMITY INSTITUTE OF INFORMATION TECHNOLOGY

WEEKLY PROGRESS REPORT
FOR THE WEEK COMMENCING (18/05/2026-24/05/2026)

WPR No: 02

Program: MCA

NAME: DHRUV DHAYAL

ENROLLMENT NO.- A010145025215

SEMESTER: 3rd

FACULTY GUIDE NAME: Dr. Garima Bohra

PROJECT TITLE: RADMAP – GPS Enabled Radiation Mapping Prototype using Raspberry Pi

TARGETS FOR THE WEEK:

1. Study the Safecast Radiation Mapping platform and its features.
2. Prepare and upload the radiation dataset on Kaggle.
3. Develop GIS-based radiation visualization using interactive maps.
4. Implement a machine learning model for radiation prediction.

ACHIEVEMENTS OF THE WEEK:

1. Successfully created and uploaded the Amity University Noida radiation dataset to Kaggle.
2. Developed interactive radiation maps and heatmap 2D,3D visualizations using Folium.
3. Performed data preprocessing, validation, and exploratory data analysis.
4. Built and tested a Random Forest model for radiation level prediction.

FUTURE WORK PLANS:

1. Expand the dataset with additional radiation monitoring locations and parameters.
2. Develop an advanced dashboard with Plotly and Streamlit for better visualization.
3. Implement anomaly detection and advanced machine learning algorithms.
4. Prepare project documentation, GitHub repository, and research paper for publication.